Semester: 5th semester
Subject Name:- CI .& Code:-CS-3013
Branch (s): - CSE, IT, CSSE

# AUTUMN MID SEMESTER EXAMINATION-2023

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject Name: Computational Intelligence
[Subject Code: CS 3013]

**Time: 1 1/2 Hours** **Full Mark: 40**

***Answer any four Questions including Question No. 1 which is compulsory.***
***The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.***

1. Answer all the questions. [ 2 x 5 ]

   a) Differentiate hard computing and soft computing?

   b) Consider a fully connected neural network consisting of two hidden layers, each containing 5 neurons. The input to this network is represented by a 4D feature vector and is designated for a classification task with 3 classes. Calculate the number of connection weights and biases in the network.

   c) How do you describe ReLU non-linearity function?
      a) Continuous and Non-differentiable
      b) Discontinuous and Non-differentiable
      c) Continuous and Differentiable
      d) Discontinuous and Differentiable

   d) Given input signals, x1=-0.11, x2 = 0.22, desired response, d= -1, synaptic weights, w1 = -0.17, w2 = 0.53, learning rate = 0.65, make necessary correction to the weights using LMS weight update rule.

   e) What factors influence the number of epochs needed to achieve convergence in a Perceptron training process?

1. What is linearly separable problem? Discuss the rules for adapting synaptic weights of perceptron. Write down the Perceptron learning algorithm and explain each step. Find the new weights after epoch-1 to classify OR function with bipolar input and targets. Assume initial weight w1=w2=b=0. learning rate = 1 and threshold = 0.

   [10 Marks ]

2. Describe a step-by-step manual numerical execution of the MADALINE (Multiple Adaline) neural network for the implementation of the XOR function using MR-I training algorithm. Initialize weights and bias as follows. The weights for first hidden neuron: $w_{10} = 0.2$, $w_{11} = 0.3$, and for second hidden neuron: $W_{12} = 0.2$, $W_{20} = 0.3$, $W_{21} = 0.2$, $W_{22} = 0.1$ with learning rate parameter $\eta = 0.5$.

   [ 10 Marks ]

4. Draw Multilayer Perceptron feed-forward neural network with 2 inputs, 2 hidden and 2 output neurons along with bias. Using Back-propagation algorithm, calculate local gradients of both the hidden layer $(\delta_1^1, \delta_2^1)$ and output layer neurons $(\delta_1^2, \delta_2^2)$. Utilize the given weight matrices for the connections from the input to the hidden layer and from the hidden to the output layer and input vector and the corresponding output vector from category 1. Assume learning rate as 0.1.

$$W^1 = \begin{bmatrix} 0.5 & 1.5 & 0.8 \\ 0.8 & 0.2 & -1.6 \end{bmatrix} \quad W^2 = \begin{bmatrix} 0.9 & -1.7 & 1.6 \\ 1.2 & 2.1 & -0.2 \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} 0.7 \\ 1.2 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

[ 10 Marks ]

5. Highlight the key differences between a Multilayer Perceptron (MLP) and Radial Basis Function (RBF) feed-forward neural networks.

[ 10 Marks ]

*** Best of Luck ***